

Professional Summary	Data and AI/ML professional with hands-on experience building scalable pipelines, machine learning systems, and NLP applications in financial services. Proficient across the modern data stack, Databricks, Snowflake, dbt, PySpark, AWS, and GCP, with a track record of delivering production-grade, cloud-native data workflows that reduce data preparation time and accelerate model development. Experienced in building analytics-ready datasets, Power BI dashboards, and integrating LLM APIs into production systems.
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Experience	AI Cloud Solutions Aug 2024 - Present Data Engineer - Designed and deployed unified batch and real-time data pipelines on Databricks and AWS (S3, EC2, Redshift), including real-time Kafka ingestion pipelines, processing structured and semi-structured transaction data from multiple sources and reducing data preparation time by 30%. - Built and optimized PySpark jobs on Databricks using Delta Lake for ACID-compliant data storage, implementing partitioning and parallel processing strategies that reduced pipeline latency and improved throughput for downstream analytics and ML workflows. - Developed modular data transformation layers using dbt on Snowflake, building reusable fact and dimension table models that standardized reporting logic and reduced redundant SQL across analytics teams, and managed data governance and lineage tracking via Databricks Unity Catalog. - Leveraged GCP BigQuery for cross-platform analytics workflows, integrating large-scale datasets alongside AWS infrastructure to support multi-cloud data environments. - Engineered data quality monitoring frameworks including reconciliation scripts and validation checks to ensure integrity and accuracy across high-impact reporting and ML pipelines. - Engineered behavioral and transactional feature pipelines generating signals for predictive models, including transaction velocity, rolling aggregations, and anomaly indicators, reducing model preparation time by 20%. - Built and maintained interactive Power BI dashboards connected to Snowflake, enabling business stakeholders to monitor financial KPIs and transaction trends in real time, reducing manual reporting hours by 40%. - Developed FastAPI endpoints to expose model outputs and pipeline results for real-time consumption by downstream applications, with average prediction retrieval latency under 200ms. - Containerized pipeline services and ML APIs using Docker and Kubernetes with automated CI/CD via Jenkins and GitHub, enabling scalable deployment and minimizing production errors. FoxFire Technologies Jan 2021 - Dec 2021 Data Analyst Intern - Performed exploratory data analysis using Python (pandas, NumPy) and SQL on CRM and ERP datasets, uncovering trends and correlations that helped senior management prioritize key business drivers and make data-informed decisions. - Developed and optimized complex SQL queries to extract, join, and transform data across multiple operational systems, reducing manual reporting time and improving data accessibility for business stakeholders. - Applied statistical techniques including hypothesis testing and anomaly detection to validate data quality across reporting pipelines, ensuring reliability of analytical outputs used in business decisions. - Designed and maintained interactive dashboards in Tableau and Power BI to monitor KPIs in real time, enabling stakeholders to track performance metrics and act faster on emerging trends.
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Projects	Fraud Detection Data Pipeline & ML System - Engineered an end-to-end fraud detection pipeline processing 284,807 real anonymized credit card transactions using PySpark for distributed ETL, mirroring production-scale data engineering architecture used in financial institutions. - Designed and implemented 5 behavioral features including log-normalized transaction amount, time-of-day extraction, rolling transaction velocity, rolling average spend, and high-value flags, transforming raw anonymized data into actionable fraud signals. - Addressed severe class imbalance (0.17% fraud rate) using class weighting with a 578:1 penalty ratio, achieving 0.98 ROC-AUC, 83% fraud recall, and 77% precision on a held-out test set across Random Forest and XGBoost classifiers.
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- Deployed a real-time prediction API via FastAPI returning fraud probability scores in under 200 milliseconds, containerized with Docker for consistent behavior across development and production environments.

LLM Powered Text Analytics & Summarization System

- Built an end-to-end NLP pipeline using BART for abstractive summarization, FinBERT for domain-specific financial sentiment analysis, and YAKE for statistical keyword extraction, automatically processing earnings call transcripts for 5 major companies and correctly identifying Apple as the only negative report at 94.88% confidence based on China revenue decline language.
- Engineered a historical market event detection system that ingested 28,877 daily stock price records spanning 10 years, flagging 1,849 major market events defined as single-day price moves exceeding 5%, and computing 1-day, 5-day, and 30-day post-earnings price reactions to quantify how markets respond to earnings sentiment signals.
- Extended the NLP pipeline to cover 10 companies by building a yfinance ingestion layer processing 454 quarterly earnings records across large-cap and mid-cap technology companies, linking sentiment outputs directly to corresponding market reaction windows for correlation analysis.
- Trained a GradientBoosting classifier on 435 labeled historical samples to predict next-quarter earnings sentiment and a regression model to forecast 30-day post-earnings price reactions, using volatility, momentum, and EPS surprise as engineered features.

AI Companion - Mobile Application

- Built a cross-platform mobile and web application using React Native and Expo, integrating the Anthropic Claude API to power emotionally intelligent conversations with a custom persona and adaptive response system.
- Designed a semantic memory system storing conversation-derived facts as 1,536-dimensional vector embeddings via OpenAI's text-embedding-3-small model in PostgreSQL with pgvector, enabling cosine similarity search to surface contextually relevant memories in real time rather than relying on keyword matching.
- Engineered a three-layer emotion detection pipeline combining behavioral signal extraction, analyzing message length trends, punctuation patterns, hedge word frequency, and time-of-day signals, with few-shot retrieval from 1,038 labeled examples sourced from Google's GoEmotions and Meta's EmpatheticDialogues datasets, and LLM synthesis via Claude API returning structured JSON emotion objects for downstream processing.
- Implemented a mood-adaptive UI theming system in React Native that dynamically transitions color palettes across 12 distinct emotional states in real time based on sentiment inference returned from the backend, creating a visually responsive experience that mirrors the user's emotional context.
- Integrated third-party APIs including OpenWeatherMap and NewsAPI to passively enrich user context, incorporating weather conditions, local events, and news sentiment into conversation analysis to identify environmental and situational factors influencing mood trends over time.
- Implemented proactive engagement features using node-cron scheduling, detecting trigger patterns in conversation such as upcoming interviews, pending decisions, and unresolved conflicts, and automatically scheduling contextual follow-up messages at appropriate times to support the user's emotional continuity.
- Enforced data isolation via Supabase row-level security policies and built a crisis detection layer that intercepts responses before LLM processing when high-risk emotional signals are present, prioritizing user safety before generating any output.

Technical Skills

- **Programming & Scripting:** Python, SQL, R, Java, C/C++, Javascript
- **Data Engineering:** Apache Spark, PySpark, Airflow, Kafka, ELT/ETL Pipelines, Snowflake, Databricks, Data Modeling, Data Warehousing, Delta Lake, dbt, PostgreSQL
- **Cloud & Infrastructure:** AWS (S3, EC2, IAM, Redshift), Kubernetes, Docker, Azure, Supabase, GCP (BigQuery, Cloud Storage), Terraform
- **APIs & Backend:** FastAPI, REST APIs, Node.js, Express
- **Analytics & Machine Learning:** NumPy, pandas, scikit-learn, Feature Engineering, Statistical Methods, Model Evaluation, Supervised Learning, NLP, pgvector, OpenAI Embeddings, Few-Shot Classification
- **Visualization & Tools:** Power BI, Tableau, Looker, Git, GitHub, Jenkins, CI/CD
- **Data Governance & Quality:** Databricks Unity Catalog, Data Lineage, Metadata Management, Data Quality Frameworks

Education

Florida International University
Master's, Data Science & Artificial Intelligence

2022 - 2024
Miami, FL